

```
!pip uninstall -y transformers
!pip install -U transformers accelerate bitsandbytes sentencepiece
```

```
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (from transformers) (0.1.99)
Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.11.0+cu128)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (8.1.7)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3.16.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2024.10.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.27.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (4.12.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.0.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.4.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)) (12.8.1)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.4.5)
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Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)) (11.7.3.90)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)) (12.8.90)
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Requirement already satisfied: nvidia-cusolver-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda-bindings]<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)) (12.8.90)
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Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0.0.2)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)) (1.1.0)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (2024.12.14)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.14.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (2.18.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=2.0.0->accelerate) (3.0.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
Downloading transformers-5.11.0-py3-none-any.whl (11.1 MB)
11.1/11.1 MB 117.5 MB/s eta 0:00:00
```

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Downloading accelerate-1.14.0-py3-none-any.whl (389 kB)
389.2/389.2 kB 36.4 MB/s eta 0:00:00
Downloading bitsandbytes-0.49.2-py3-none-anylinux_2_24_x86_64.whl (60.7 MB)
60.7/60.7 MB 13.3 MB/s eta 0:00:00
```

```
Installing collected packages: bitsandbytes, accelerate, transformers
  Attempting uninstall: accelerate
    Found existing installation: accelerate 1.13.0
    Uninstalling accelerate-1.13.0:
      Successfully uninstalled accelerate-1.13.0
Successfully installed accelerate-1.14.0 bitsandbytes-0.49.2 transformers-5.11.0
```

```
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig
import pandas as pd
import re
```

```
def extract_final_num(text):
    match = re.search(r"Final\s+Answer:\s*(-?\d+\.?\d*)", text, re.IGNORECASE)
    if match:
        return match.group(1)
    nums = re.findall(r'[-?\d+\.?\d*]', text)
    if nums:
        return nums[-1]
    return None
```

```
test_df = pd.read_csv("unified_svamp_test.csv").head(50)
```

```
import os
os.environ['LD_LIBRARY_PATH'] += ':/usr/local/cuda-13.0/lib64'
```

```
from google.colab import userdata
from huggingface_hub import login
```

```
hf_token = userdata.get('HF_TOKEN')
login(hf_token)
print("Authenticated successfully!")
```

Authenticated successfully!

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig
model_id = "google/gemma-2b-it"
bnb_config = BitsAndBytesConfig(
    load_in_4bit = True,
    bnb_4bit_quant_type = "nf4",
    bnb_4bit_compute_dtype=torch.float16
)
```

```
tokenizer = AutoTokenizer.from_pretrained(model_id)
```

```
config.json: 100%                               627/627 [00:00<00:00, 60.5kB/s]
tokenizer_config.json: 100%                     34.2k/34.2k [00:00<00:00, 2.69MB/s]
tokenizer.json: 100%                             17.5M/17.5M [00:01<00:00, 15.1MB/s]
special_tokens_map.json: 100%                   636/636 [00:00<00:00, 14.0kB/s]
```

```
print("Downloading the model")
model = AutoModelForCausalLM.from_pretrained(model_id, quantization_config=bnb_config, device_map="auto")
print("Success! The engine is running")
```

```
Downloading the model
model.safetensors.index.json: 100%              13.5k/13.5k [00:00<00:00, 402kB/s]

Download complete: 100%    5.01G/5.01G [03:41<00:00, 21.7MB/s]

Fetching 2 files: 100%                2/2 [03:41<00:00, 221.45s/it]

Loading weights: 100%              164/164 [00:20<00:00, 8.97it/s]

generation_config.json: 100%          137/137 [00:00<00:00, 13.7kB/s]

Success! The engine is running
```

```
results = []
correct_count = 0
total_count = len(test_df)
```

```
print("\nStarting Baseline Evaluation-->")
for index, row in test_df.iterrows():
    question = row['question']
    ground_truth = str(row['answer']).strip()
    prompt = f"You are an expert math tutor. Question: {question}\nAnswer the question step-by-step. You must strictly end your inputs = tokenizer(prompt, return_tensors='pt').to('cuda')
    outputs = model.generate(
        **inputs,
        max_new_tokens=300,
        temperature=0.1,
        do_sample=True,
        repetition_penalty=1.15
    )
    response = tokenizer.decode(outputs[0][inputs['input_ids'].shape[-1]:], skip_special_tokens=True)
    extracted_answer = extract_final_num(response)

    is_correct = False
    if extracted_answer is not None:
        try:
            # Compare as floats to handle "16" vs "16.0" safely
            if float(extracted_answer) == float(ground_truth):
```

```

        is_correct = True
        correct_count += 1
    except ValueError:
        pass

    print(f"\n--- Problem {index + 1} ---")
    print(f"Question: {question}")
    print(f"Model Output:\n{response.strip()}")
    print(f"Extracted Answer: {extracted_answer}")
    print(f"Ground Truth Answer: {ground_truth}")
    print(f"Correct: {is_correct}")
    print("-" * 30)

```

--- Problem 48 ---

Question: After resting they decided to go for a swim. The depth of the water is 2 times Dean's height. Dean is 8 feet shorter than Ron. How deep is the water?
Model Output:
16

****Explanation:****

Let x be Dean's height. Then Ron's height is x + 14.

Since the depth is 2 times Dean's height, we have:

$$x = 2(x + 14)$$

****Solving for x:****

$$x - 28 = 2x - 28$$

$$x - 2x = 28 + 28$$

$$0x = 56$$

$$x = 56 / 0$$

This is not a valid solution, which means that Ron cannot be 8 feet tall. Therefore, the depth of the water is 16 feet.

Extracted Answer: 16
Ground Truth Answer: 12
Correct: False

--- Problem 49 ---

Question: Every day Ryan spends 6 hours on learning English and 7 hours on learning Chinese. If he learns for 5 days How many hours will he have spent learning?
Model Output:
30 hours.

****Explanation:****

- * Ryan spends 6 hours on learning English each day.
- * He spends 7 hours on learning Chinese each day.
- * In 5 days, he will spend 6 hours on learning English and 7 hours on learning Chinese, a total of 13 hours.
- * Therefore, the total number of hours he spends on learning English and Chinese in all is 13 hours.

Extracted Answer: 13
Ground Truth Answer: 65
Correct: False

--- Problem 50 ---

Question: Jack received 10 emails in the morning, 5 emails in the afternoon and 4 emails in the evening. How many more emails did he receive in the morning than in the afternoon?
Model Output:
5

Final Answer: 5
Extracted Answer: 5
Ground Truth Answer: 1
Correct: False

```

accuracy = (correct_count / total_count) * 100
print(f"FINAL BASELINE ACCURACY: {accuracy:.2f}% ({correct_count}/{total_count})")

```

FINAL BASELINE ACCURACY: 76.00% (38/50)